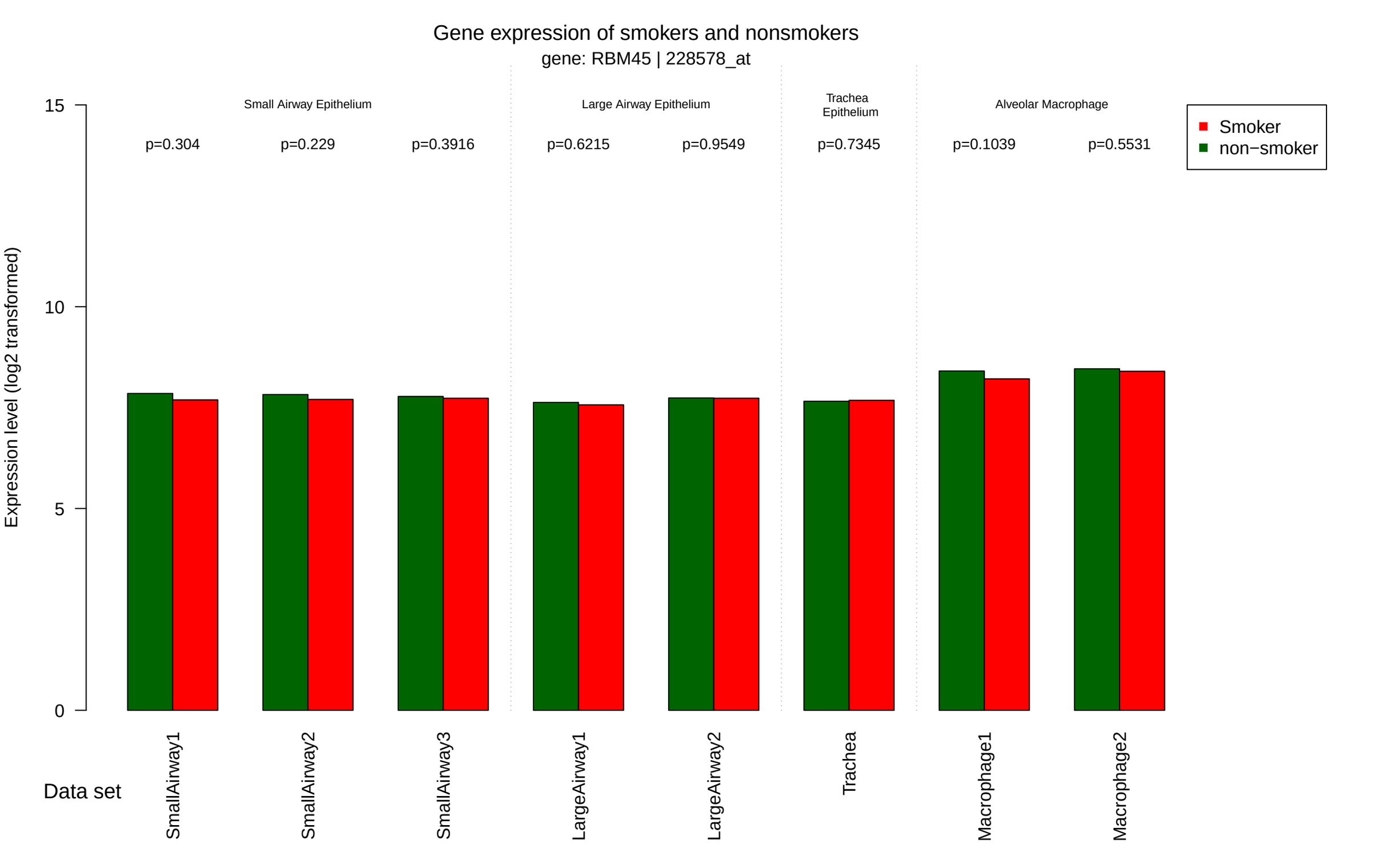
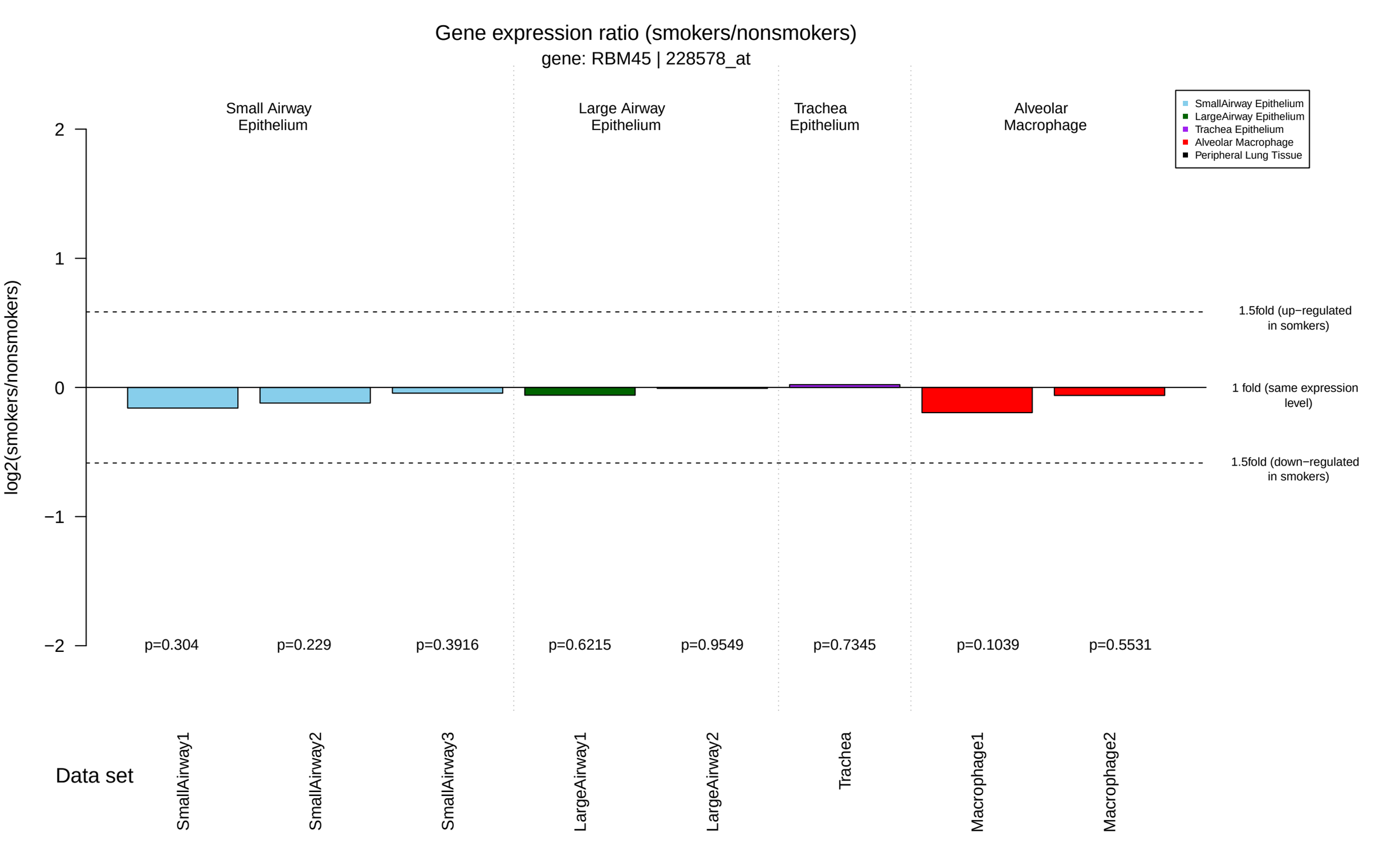
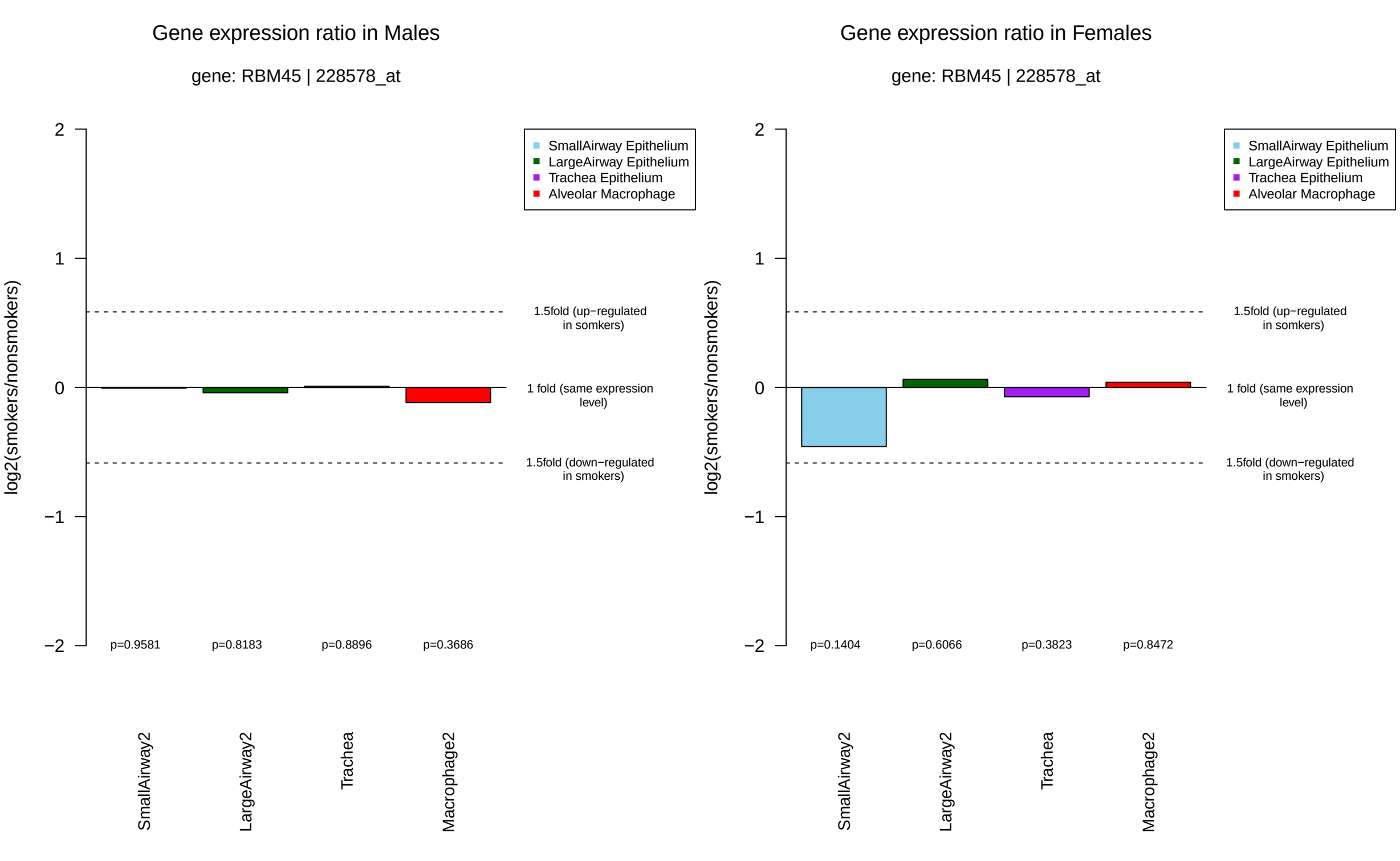
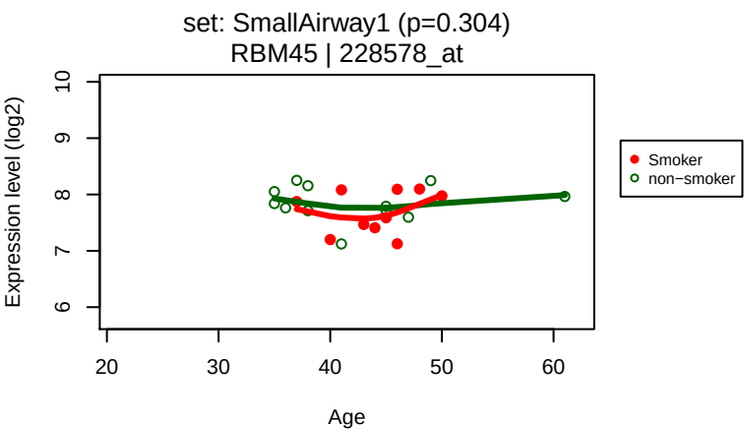
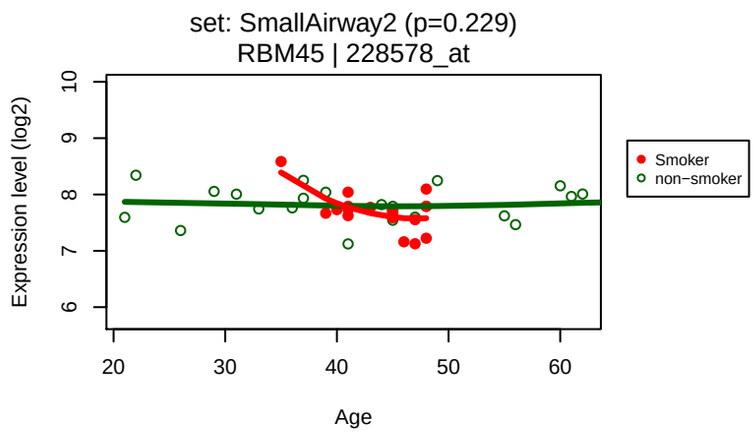
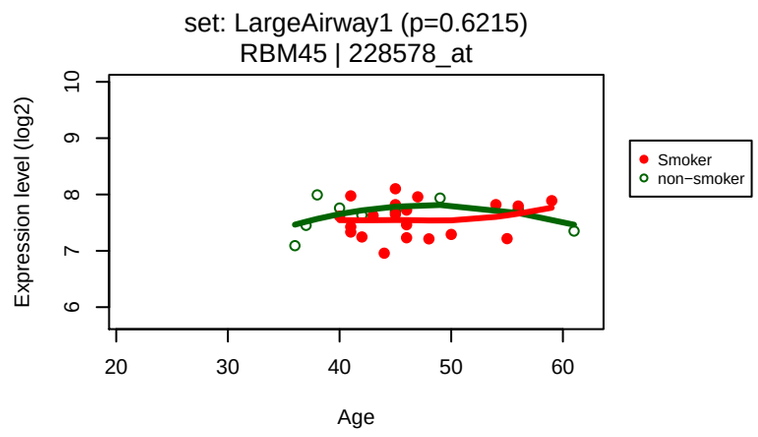

Gene expression of smokers and nonsmokers
gene: RBM45 | 228578_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.304
p=0.229
p=0.3916
p=0.6215
p=0.9549
p=0.7345
p=0.1039
p=0.5531
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

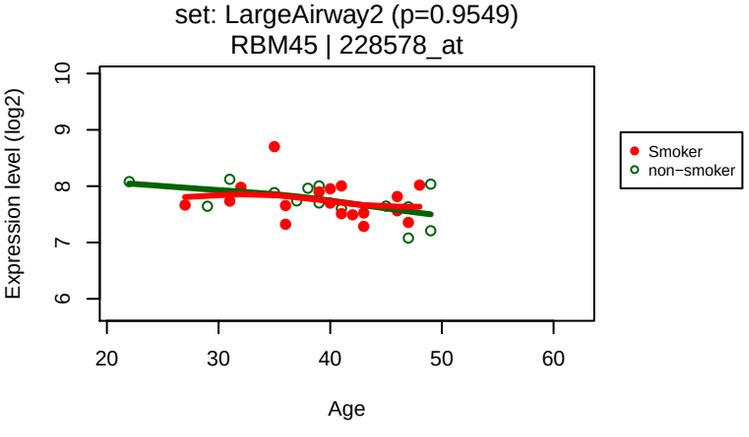

Gene expression ratio (smokers/nonsmokers)
gene: RBM45 | 228578_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.304
p=0.229
p=0.3916
p=0.6215
p=0.9549
p=0.7345
p=0.1039
p=0.5531
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

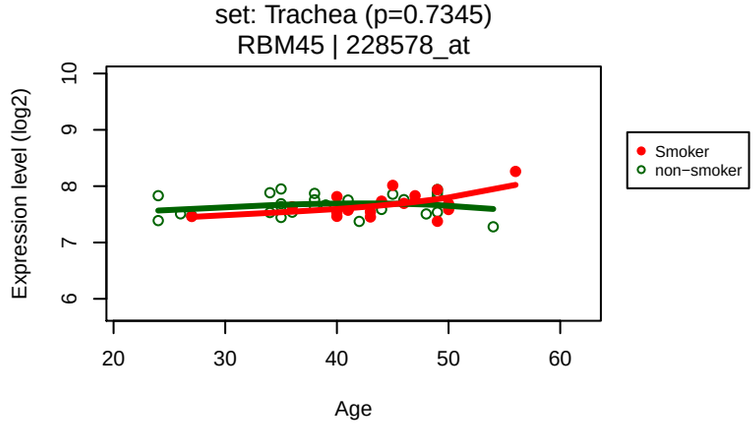

Gene expression ratio in Males
gene: RBM45 | 228578_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9581
p=0.8183
p=0.8896
p=0.3686
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RBM45 | 228578_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1404
p=0.6066
p=0.3823
p=0.8472
SmallAirway2
LargeAirway2
Trachea
Macrophage2

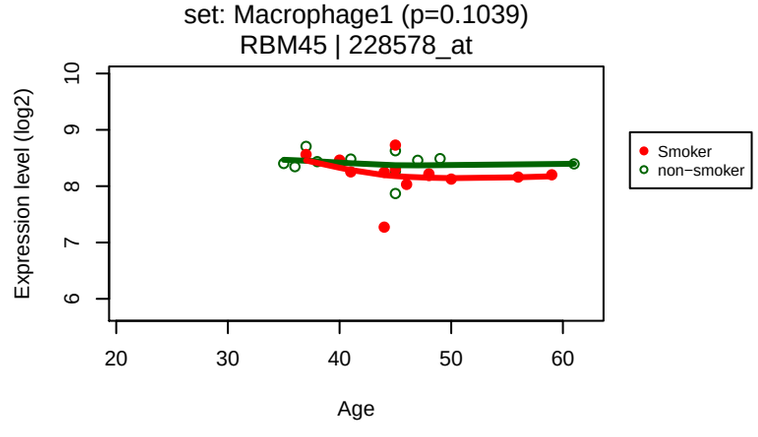

set: SmallAirway1 (p=0.304)
RBM45 | 228578_at
Expression level (log2)
Age
Smoker
non−smoker

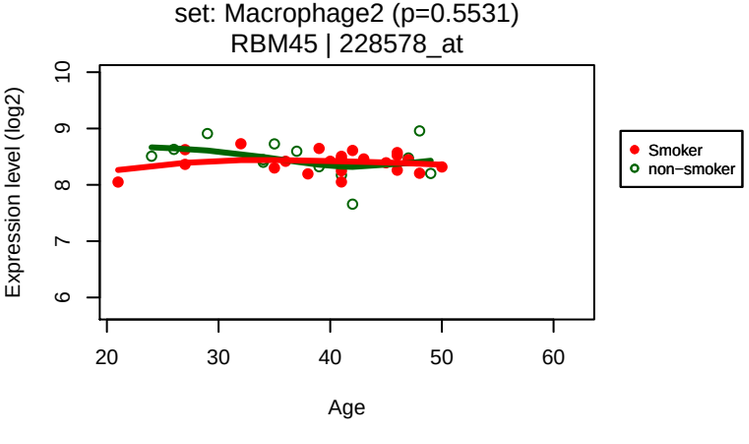

set: SmallAirway2 (p=0.229)
RBM45 | 228578_at
Expression level (log2)
Age
Smoker
non−smoker

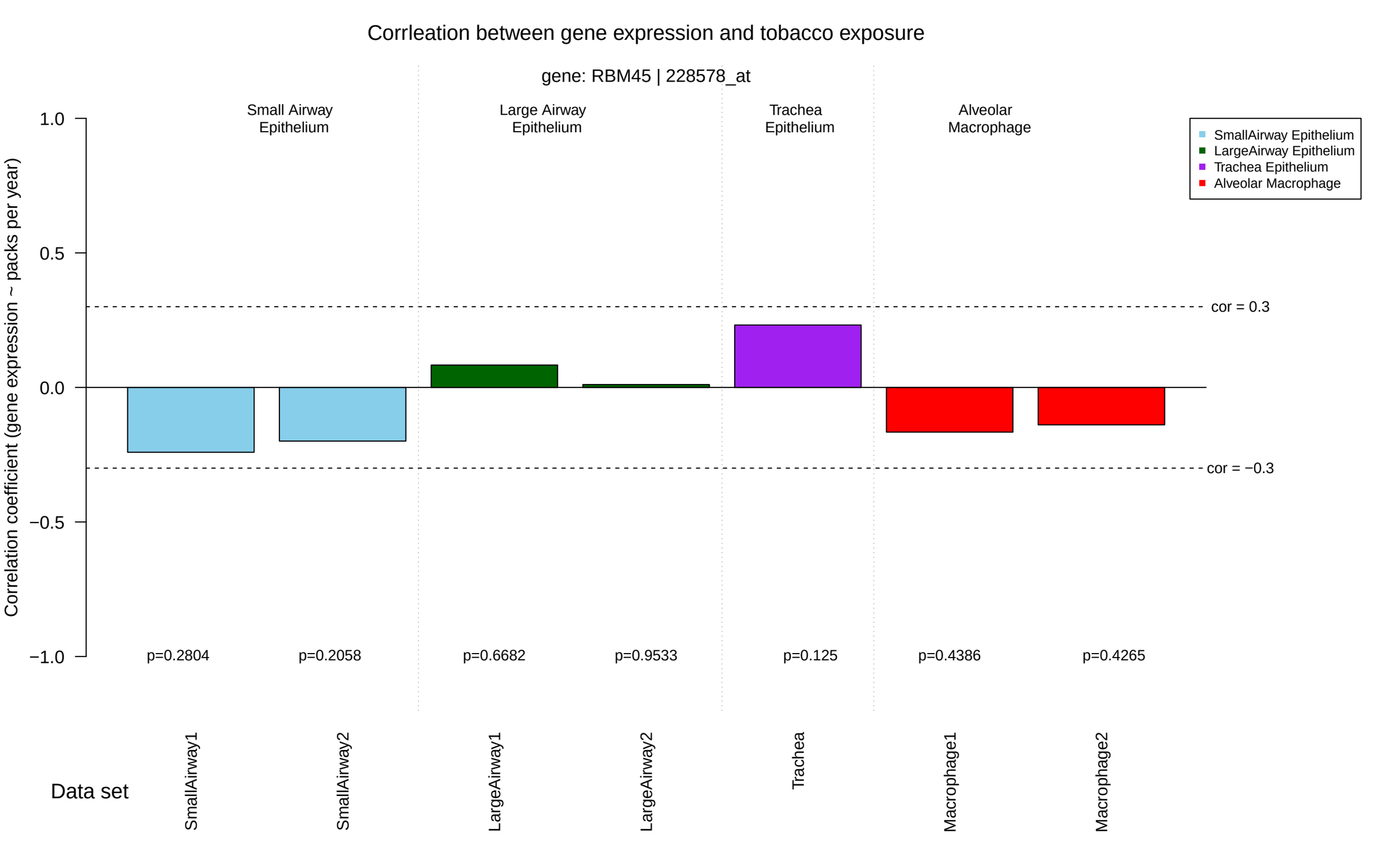

set: LargeAirway1 (p=0.6215)
RBM45 | 228578_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9549)
RBM45 | 228578_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7345)
RBM45 | 228578_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1039)
RBM45 | 228578_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5531)
RBM45 | 228578_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RBM45 | 228578_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2804
p=0.2058
p=0.6682
p=0.9533
p=0.125
p=0.4386
p=0.4265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2